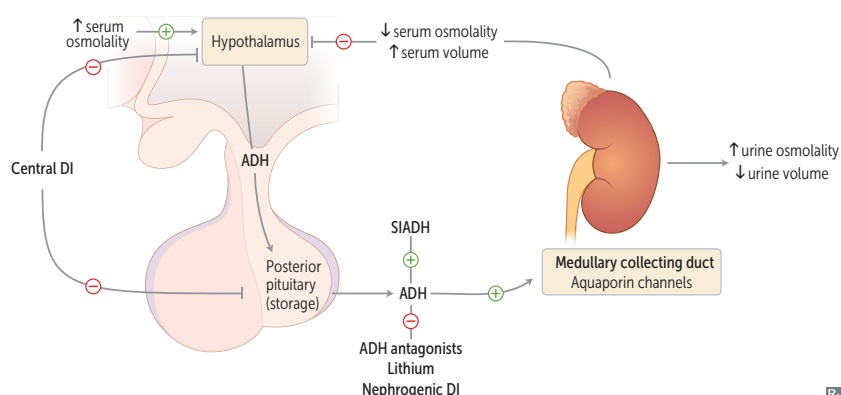


► ENDOCRINE—PATHOLOGY

Syndrome of inappropriate antidiuretic hormone secretion

Characterized by excessive free water retention, euvoletic hyponatremia with continued urinary Na^+ excretion, urine osmolality $>$ serum osmolality.

Body responds to water retention with $\downarrow$ aldosterone and $\uparrow$ ANP and BNP $\rightarrow$ $\uparrow$ urinary Na^+ secretion $\rightarrow$ normalization of extracellular fluid volume $\rightarrow$ euvoletic hyponatremia.

Treatment: fluid restriction (first line), salt tablets, IV hypertonic saline, diuretics, ADH antagonists (eg, conivaptan, tolvaptan, demeclocycline).

SIADH causes include (**HEELD**-up water):

- **H**ead trauma/CNS disorders
- **E**ctopic ADH (eg, small cell lung cancer)
- **E**xogenous hormones (eg, vasopressin, desmopressin, oxytocin)
- **L**ung disease
- **D**rugs (eg, SSRIs, carbamazepine, cyclophosphamide)

Primary polydipsia and diabetes insipidus

Characterized by the production of large amounts of dilute urine +/- thirst. Urine specific gravity < 1.006 . Urine osmolality usually < 300 mOsm/kg. Central DI may be transient if damage is below hypothalamic median eminence or in the posterior pituitary (ADH in hypothalamus can still be secreted systemically via portal capillaries in median eminence). Central DI, also known as arginine vasopressin deficiency; nephrogenic DI, also known as arginine vasopressin resistance.

	Primary polydipsia	Central DI	Nephrogenic DI
DEFINITION	Excessive water intake	$\downarrow$ ADH release	ADH resistance
CAUSES	Psychiatric illnesses, hypothalamic lesions affecting thirst center	Idiopathic, brain injury (trauma, hypoxia, tumor, surgery, infiltrative diseases)	Hereditary (V_2 -receptor mutation), drugs (eg, lithium, demeclocycline), hypercalcemia, hypokalemia
SERUM OSMOLALITY	$\downarrow$	$\uparrow$	$\uparrow$
ADH LEVEL	$\downarrow$ or normal	$\downarrow$	Normal or $\uparrow$
WATER RESTRICTION^a	Significant $\uparrow$ in urine osmolality (> 700 mOsm/kg)	No change or slight $\uparrow$ in urine osmolality	No change or slight $\uparrow$ in urine osmolality
DESMOPRESSIN ADMINISTRATION^b	—	Significant $\uparrow$ in urine osmolality ($> 50\%$)	Minimal change in urine osmolality
TREATMENT	Water restriction	Desmopressin (DDAVP)	Manage the underlying cause; low-solute diet, HCTZ, amiloride, indomethacin

^aNo water intake for 2–3 hours followed by hourly measurements of urine volume and osmolality as well as plasma Na^+ concentration and osmolality.

^bDesmopressin (ADH analog) is administered if serum osmolality > 295 – 300 mOsm/kg, plasma $\text{Na}^+ \geq 145$ mEq/L, or urine osmolality does not increase despite $\uparrow$ plasma osmolality.

Hypopituitarism

Undersecretion of pituitary hormones due to:

- Hypothalamic disease (ie, mass lesion, radiation, infiltrative lesions)
- Pituitary surgery
- Pituitary disorders (eg, mass effect due to nonsecreting pituitary adenoma, craniopharyngioma)
- Empty sella
- **Sheehan syndrome**—ischemic infarct of pituitary following severe postpartum hemorrhage; pregnancy-induced pituitary growth → ↑ susceptibility to hypoperfusion. Usually presents with failure to lactate, amenorrhea, cold intolerance (anterior pituitary hormones mainly affected).

Treatment: hormone replacement therapy (glucocorticoids, thyroxine, sex steroids, human growth hormone)

Acromegaly

Excess GH in adults. Typically caused by pituitary adenoma.

FINDINGS

Large tongue with deep furrows, frontal bossing, coarsening of facial features with aging **A**, deep voice, diaphoresis (excessive sweating), hypertrophic arthropathy, impaired glucose tolerance (insulin resistance), HTN, LVH, HFpEF (most common cause of death).

↑ GH in children → gigantism (↑ linear bone growth due to unfused epiphysis).

Acromegaly in **adults**, gigantism in j(**g**)uniors.

DIAGNOSIS

↑ serum IGF-1 (GH levels unreliable as they fluctuate throughout the day); failure to suppress serum GH following oral glucose tolerance test; pituitary mass seen on brain MRI.

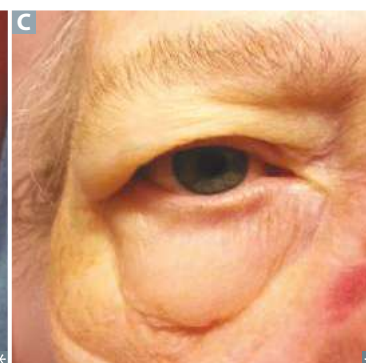


TREATMENT

Pituitary adenoma resection. If not cured, treat with octreotide (somatostatin analog), pegvisomant (GH receptor antagonist), or dopamine agonists (eg, cabergoline).

Hypothyroidism vs hyperthyroidism

	Hypothyroidism	Hyperthyroidism
METABOLIC	Cold intolerance, ↓ sweating, weight gain (↓ basal metabolic rate → ↓ calorogenesis), hyponatremia (↓ free water clearance)	Heat intolerance, ↑ sweating, weight loss (↑ synthesis of Na^+/K^+ -ATPase → ↑ basal metabolic rate → ↑ calorogenesis)
SKIN/HAIR	Dry, cool skin (due to ↓ blood flow); coarse, brittle hair; diffuse alopecia; brittle nails; puffy facies and generalized nonpitting edema (myxedema) due to ↑ GAGs in interstitial spaces → ↑ osmotic pressure → water retention	Warm, moist skin (due to vasodilation); fine hair; onycholysis (A); pretibial myxedema in Graves disease B
OCULAR	Periorbital edema C	Ophthalmopathy in Graves disease (including periorbital edema, exophthalmos), lid lag/retraction (↑ sympathetic stimulation of superior tarsal muscle)
GASTROINTESTINAL	Constipation (↓ GI motility), ↓ appetite	Hyperdefecation/diarrhea (↑ GI motility), ↑ appetite
MUSCULOSKELETAL	Hypothyroid myopathy (proximal weakness, ↑ CK), carpal tunnel syndrome, myoedema (small lump rising on the surface of a muscle when struck with a hammer)	Thyrotoxic myopathy (proximal weakness, normal CK), osteoporosis/↑ fracture rate (T_3 stimulates osteoclast differentiation, ↑ bone resorption, and release of calcium; may also affect osteoblasts)
REPRODUCTIVE	Abnormal uterine bleeding, ↓ libido, infertility	Abnormal uterine bleeding, gynecomastia, ↓ libido, infertility
NEUROPSYCHIATRIC	Hypoactivity, lethargy, fatigue, weakness, depressed mood, ↓ reflexes (delayed/slow relaxing)	Hyperactivity, restlessness, anxiety, insomnia, fine tremors (due to ↑ β -adrenergic activity), ↑ reflexes (brisk)
CARDIOVASCULAR	Bradycardia, dyspnea on exertion (↓ cardiac output)	Tachycardia, palpitations, dyspnea, arrhythmias (eg, atrial fibrillation), chest pain and systolic HTN due to ↑ number and sensitivity of β -adrenergic receptors, ↑ expression of cardiac sarcolemmal ATPase and ↓ expression of phospholamban
LABS	↑ TSH (if 1°) ↓ free T_4 Hypercholesterolemia (due to ↓ LDL receptor expression)	↓ TSH (if 1°) ↑ free T_3 and T_4 ↓ LDL, HDL, and total cholesterol



Hypothyroidism**Hashimoto thyroiditis**

Also called chronic autoimmune thyroiditis. Most common cause of hypothyroidism in iodine-sufficient regions. Associated with HLA-DR3 (differs by ethnicity), ↑ risk of primary thyroid lymphoma (typically diffuse large B-cell lymphoma).

Findings: moderately enlarged, **nontender** thyroid. May be preceded by transient hyperthyroid state (“Hashitoxicosis”) due to follicular rupture and thyroid hormone release.

Serology: ⊕ antithyroid peroxidase (antimicrosomal) and antithyroglobulin antibodies.

Histology: Hürthle cells **A**, lymphoid aggregates with germinal centers **B**.

Postpartum thyroiditis—mild, self-limited variant of Hashimoto thyroiditis arising < 1 year after delivery. Natural history: hyperthyroidism (within 3 months of birth) → hypothyroidism → recovery/euthyroid state.

Subacute granulomatous thyroiditis

Also called de Quervain thyroiditis. Usually, a self-limited disease. Natural history: transient hyperthyroidism → euthyroid state → hypothyroidism → euthyroid state. Often preceded by viral infection.

Findings: ↑ ESR, jaw pain, very **tender** thyroid (de Quervain is associated with **pain**).

Histology: granulomatous inflammation **C**.

Riedel thyroiditis

Also called invasive fibrous thyroiditis. May occur as part of IgG₄-related disease spectrum (eg, autoimmune pancreatitis, retroperitoneal fibrosis, noninfectious aortitis). Hypothyroidism occurs in 1/3 of patients. Fibrosis may extend to local structures (eg, trachea, esophagus), mimicking anaplastic carcinoma.

Findings: slowly enlarging, hard (rocklike), fixed, **nontender** thyroid.

Histology: thyroid replaced by fibrous tissue and inflammatory infiltrate **D**.

Congenital hypothyroidism

Formerly called cretinism. Most commonly caused by thyroid dysgenesis (abnormal thyroid gland development; eg, agenesis, hypoplasia, ectopy) or dyshormonogenesis (abnormal thyroid hormone synthesis; eg, mutations in thyroid peroxidase) in iodine-sufficient regions.

Findings (**6 P**'s): **p**ot-bellied, **p**ale, **p**uffy-faced child **E** with **p**rotruding umbilicus, **p**rotuberant tongue **F**, and **p**oor brain development.

Other causes

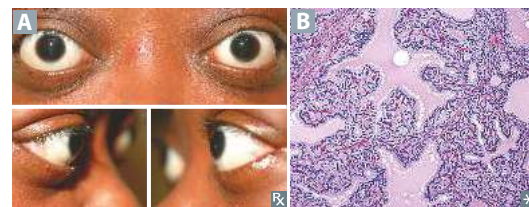
Iodine deficiency (most common cause worldwide; typically presents with goiter **G**), iodine excess (Wolff-Chaikoff effect), drugs (eg, amiodarone, lithium), nonthyroidal illness syndrome (also called euthyroid sick syndrome; ↓ T₃ with normal/↓ T₄ and TSH in critically ill patients).



Hyperthyroidism**Graves disease**

Most common cause of hyperthyroidism. Thyroid-stimulating immunoglobulin (IgG, can cause transient neonatal hyperthyroidism; type II hypersensitivity) stimulates TSH receptors on thyroid (hyperthyroidism, diffuse goiter), dermal fibroblasts (pretibial myxedema), and orbital fibroblasts (Graves orbitopathy; treat with glucocorticoids). Activation of T-cells → lymphocytic infiltration of retroorbital space → ↑ cytokines (eg, TNF- α , IFN- γ) → ↑ fibroblast secretion of hydrophilic GAGs → ↑ osmotic muscle swelling, muscle inflammation, and adipocyte count → exophthalmos **A**. Often presents during stress (eg, pregnancy). Associated with HLA-DR3 and HLA-B8.

Histology: tall, crowded follicular epithelial cells; scalloped colloid **B**.

**Toxic multinodular goiter**

Focal patches of hyperfunctioning follicular cells distended with colloid working independently of TSH (due to TSH receptor mutations in 60% of cases). ↑ release of T₃ and T₄. Hot nodules (hyperfunctioning nodules visualized on radioactive iodine scan) are rarely malignant.

Thyroid storm

Uncommon but serious complication that occurs when hyperthyroidism is incompletely treated/untreated and then significantly worsens in the setting of acute stress such as infection, trauma, surgery. Presents with agitation, delirium, fever, diarrhea, coma, and tachyarrhythmia (cause of death). May see ↑ LFTs. Treat with the **4 P's**: β -blockers (eg, **p**ropranolol), **p**ropylthiouracil, glucocorticoids (eg, **p**rednisolone), **p**otassium iodide (Lugol iodine). Iodide load → ↓ T₄ synthesis → Wolff-Chaikoff effect.

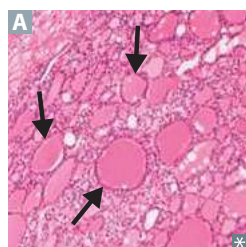
Other causes

Exogenous thyrotoxicosis (excessive intake of thyroid hormone; suspect in patient trying to lose weight), ectopic thyroid production (struma ovarii), contrast-induced thyroiditis (Jod-Basedow phenomenon), drug-induced thyroiditis (amiodarone, lithium).

Causes of goiter

Smooth/diffuse: Graves disease, Hashimoto thyroiditis, iodine deficiency, TSH-secreting pituitary adenoma.

Nodular: toxic multinodular goiter, thyroid adenoma, thyroid cancer, thyroid cyst.

Thyroid adenoma

Benign solitary growth of the thyroid. Most are nonfunctional (“cold” on radioactive iodine scan), can rarely cause hyperthyroidism via autonomous thyroid hormone production (“hot” or “toxic”). Most common histology is follicular (arrows in **A**); absence of capsular or vascular invasion (unlike follicular carcinoma).

Thyroid cancer

Typically diagnosed with fine needle aspiration; treated with thyroidectomy. Complications of surgery include hypocalcemia (due to removal of parathyroid glands), transection of recurrent laryngeal nerve during ligation of inferior thyroid artery (leads to dysphagia and dysphonia [hoarseness]), and injury to the external branch of the superior laryngeal nerve during ligation of superior thyroid vascular pedicle (may lead to loss of tenor usually noticeable in professional voice users).

Papillary carcinoma

Most common. Empty-appearing nuclei with central clearing (“**Orphan Annie**” eyes) **A**, psam**Moma** bodies, nuclear grooves (**Papi** and **Moma** adopted **Orphan Annie**). ↑ risk with *RET/PTC* rearrangements and *BRAF* mutations, childhood irradiation.

Papillary carcinoma: most **p**revalent, **p**alpable lymph nodes. Good prognosis.

Follicular carcinoma

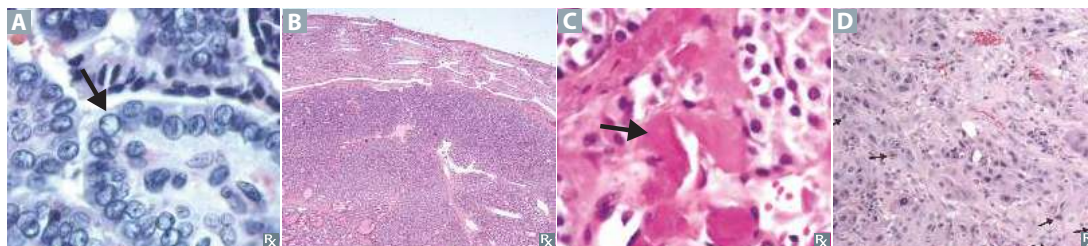
Good prognosis. Invades tumor capsule and vasculature (unlike follicular adenoma), uniform follicles **B**; hematogenous spread is common. Associated with *RAS* mutation and *PAX8-PPAR-γ* translocations. Fine needle aspiration cytology may not be able to distinguish between follicular adenoma and carcinoma.

Medullary carcinoma

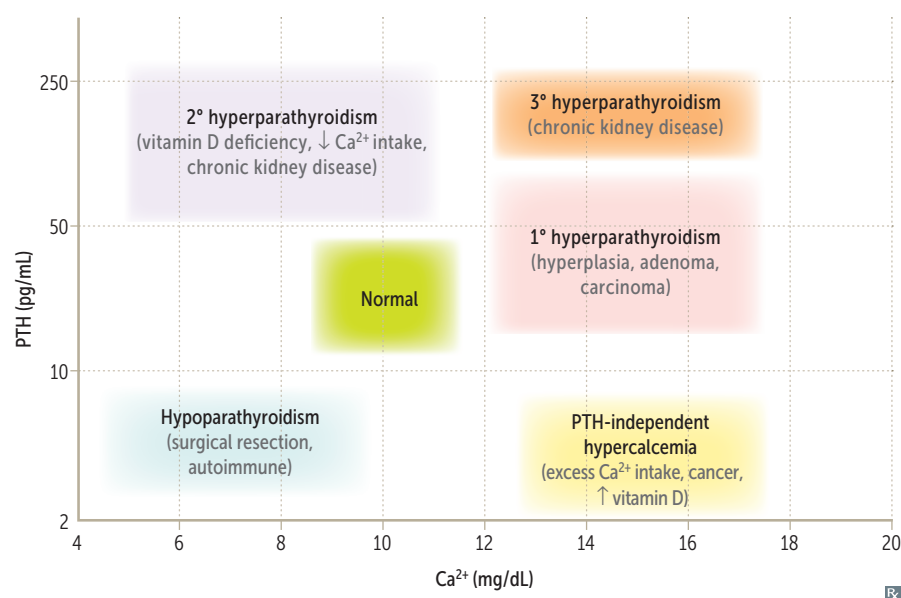
From parafollicular “**C** cells”; produces **calcitonin**, sheets of polygonal cells in an amyloid stroma **C** (stains with **C**ongo red). Associated with **MEN 2A** and **2B** (*RET* mutations; **MEN2**lary carcinoma).

**Undifferentiated/
anaplastic carcinoma**

Older patients; presents with rapidly enlarging neck mass → compressive symptoms (eg, dyspnea, dysphagia, hoarseness); very poor prognosis. Typically composed of pleomorphic cells with frequent mitoses **D**. May arise de novo, but most often arises from dedifferentiation of a preexisting follicular or papillary thyroid carcinoma.



Diagnosing parathyroid disease



Hypoparathyroidism



Due to injury to parathyroid glands or their blood supply (usually during thyroid surgery), autoimmune destruction, or DiGeorge syndrome. Findings: tetany, hypocalcemia, hyperphosphatemia.

Chvostek sign—tapping of facial nerve (tap the **C**heek) → contraction of facial muscles.

Trousseau sign—occlusion of brachial artery with BP cuff (cuff the **T**riceps) → carpal spasm.

Pseudohypoparathyroidism type 1A—autosomal dominant, maternally transmitted mutations (imprinted *GNAS* gene). *GNAS*1-inactivating mutation (coupled to PTH receptor) that encodes the G_s protein α subunit → inactivation of adenylate cyclase when PTH binds to its receptor → end-organ resistance (kidney and bone) to PTH.

Physical findings: Albright hereditary osteodystrophy (shortened 4th/5th digits **A**), short stature, round face, subcutaneous calcifications, developmental delay).

Labs: ↑ PTH, ↓ Ca²⁺, ↑ PO₄³⁻.

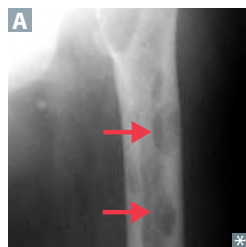
Pseudopseudohypoparathyroidism—autosomal dominant, paternally transmitted mutations (imprinted *GNAS* gene) but without end-organ resistance to PTH due to normal maternal allele maintaining renal responsiveness to PTH.

Physical findings: same as Albright hereditary osteodystrophy.

Labs: normal PTH, Ca²⁺, PO₄³⁻.

Lab values in hypocalcemic disorders

DISORDER	Ca ²⁺	PO ₄ ³⁻	PTH	ALP	25(OH) VITAMIN D	1,25(OH) ₂ VITAMIN D
Vitamin D deficiency	—/↓	—/↓	↑	↑	↓	—/↑
2° hyperparathyroidism (CKD)	↓	↑	↑	↑	—	↓
Hypoparathyroidism	↓	↑	↓	—	—	—/↓
Pseudohypoparathyroidism	↓	↑	↑	↑	—	—/↓

Hyperparathyroidism**Primary hyperparathyroidism**

Usually due to parathyroid adenoma or hyperplasia. **Hypercalcemia**, hypercalciuria (renal **stones**), polyuria (**thrones**), hypophosphatemia, ↑ PTH, ↑ ALP, ↑ urinary cAMP. Most often asymptomatic. May present with **bone** pain, weakness, constipation (**“groans”**), abdominal/flank pain (kidney stones, acute pancreatitis), neuropsychiatric disturbances (**“psychiatric overtones”**).

Osteitis fibrosa cystica—cystic **bone** spaces filled with brown fibrous tissue **A** (“brown tumor” consisting of osteoclasts and deposited hemosiderin from hemorrhages; causes bone pain). Due to ↑ PTH, classically associated with 1° (but also seen with 2°) hyperparathyroidism.

“Stones, thrones, bones, groans, and psychiatric overtones.”

Secondary hyperparathyroidism

2° hyperplasia due to ↓ Ca^{2+} absorption and/or ↑ PO_4^{3-} , most often in chronic kidney disease (causes hypovitaminosis D and hyperphosphatemia → ↓ Ca^{2+}).

Hypocalcemia, hyperphosphatemia in chronic kidney disease (vs hypophosphatemia with most other causes), ↑ ALP, ↑ PTH.

Renal osteodystrophy—renal disease → 2° and 3° hyperparathyroidism → bone lesions.

Tertiary hyperparathyroidism

Refractory (autonomous) hyperparathyroidism resulting from end-stage renal disease. ↑↑ PTH, ↑ Ca^{2+} .

Familial hypocalciuric hypercalcemia

Autosomal dominant. Defective G-coupled Ca^{2+} -sensing receptors in multiple tissues (eg, parathyroids, kidneys). Higher than normal Ca^{2+} levels required to suppress PTH. Excessive renal Ca^{2+} reabsorption → mild hypercalcemia and hypocalciuria with normal to ↑ PTH levels.

Diabetes mellitus**ACUTE MANIFESTATIONS**

Polydipsia, polyuria, polyphagia (3 P's), weight loss, DKA (type 1), hyperosmolar hyperglycemic state (type 2).

Rarely, can be caused by unopposed secretion of GH and epinephrine. Also seen in patients on glucocorticoid therapy (steroid diabetes).

CHRONIC COMPLICATIONS

Nonenzymatic glycation:

- Small vessel disease (hyaline arteriolosclerosis) → retinopathy, neuropathy, nephropathy.
- Large vessel disease (atherosclerosis) → CAD, cerebrovascular disease, peripheral vascular disease. MI is the most common cause of death.

Osmotic damage (sorbitol accumulation in organs with aldose reductase and ↓ or absent sorbitol dehydrogenase):

- Neuropathy: motor, sensory (glove and stocking distribution), autonomic degeneration (eg, GERD, gastroparesis, diabetic diarrhea).
- Cataracts.

DIAGNOSIS**TEST**

HbA_{1c}

DIAGNOSTIC CUTOFF

≥ 6.5%

NOTES

Reflects average blood glucose over prior 3 months (influenced by RBC turnover)

Fasting plasma glucose

≥ 126 mg/dL

Fasting for > 8 hours

2-hour oral glucose tolerance test

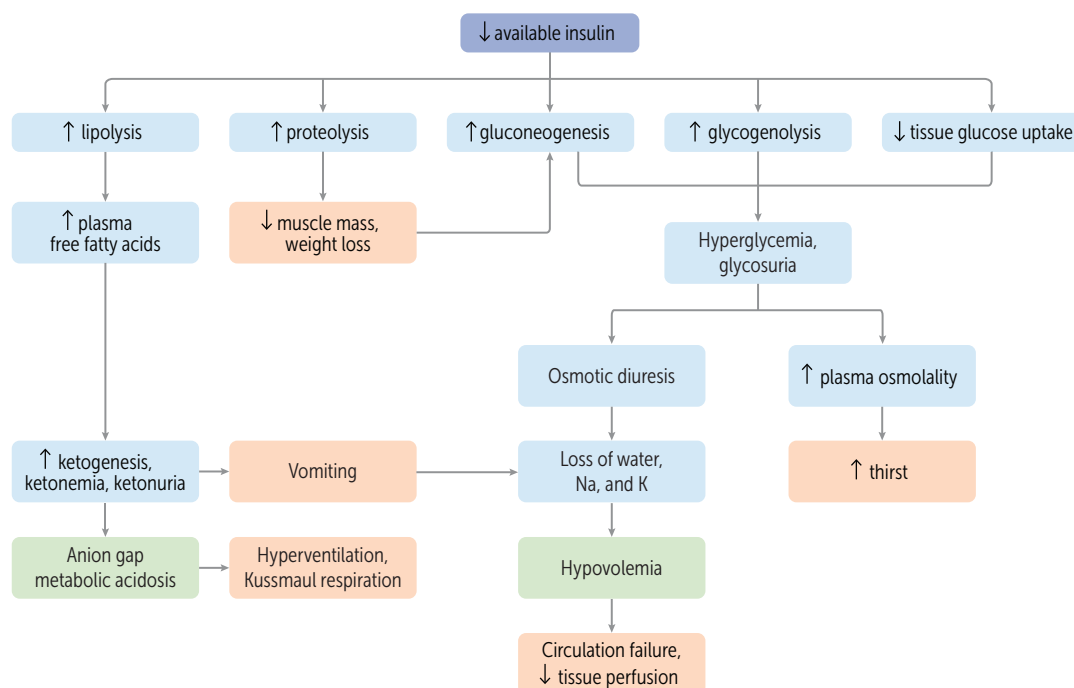
≥ 200 mg/dL

2 hours after consumption of 75 g of glucose in water

Random plasma glucose

≥ 200 mg/dL

Presence of hyperglycemic symptoms is required



Type 1 vs type 2 diabetes mellitus

	Type 1	Type 2
1° DEFECT	Autoimmune T-cell–mediated destruction of β cells (islet cell cytoplasmic +/- antiglutamic acid decarboxylase +/- insulin autoantibodies often present)	↑ resistance to insulin, progressive pancreatic β -cell failure
INSULIN NECESSARY IN TREATMENT	Always	Sometimes
AGE (EXCEPTIONS COMMON)	< 30 yr	> 40 yr
ASSOCIATION WITH OBESITY	No	Yes
GENETIC PREDISPOSITION	Relatively weak (50% concordance in identical twins), polygenic	Relatively strong (90% concordance in identical twins), polygenic
ASSOCIATION WITH HLA SYSTEM	Yes, HLA-DR4 and -DR3 (4 – 3 = type 1)	No
GLUCOSE INTOLERANCE	Severe	Mild to moderate
INSULIN SENSITIVITY	High	Low
KETOACIDOSIS	Common	Rare
β -CELL NUMBERS IN THE ISLETS	↓	Variable (with amyloid deposits)
SERUM INSULIN LEVEL	↓	↑ initially, but ↓ in advanced disease
CLASSIC SYMPTOMS ^a	Common	Sometimes
HISTOLOGY	Islet leukocytic infiltrate	Islet amyloid polypeptide deposits

^aPolydipsia, polyuria, polyphagia, weight loss

Hyperglycemic emergencies

	Diabetic ketoacidosis	Hyperosmolar hyperglycemic state
PATHOGENESIS	Insulin noncompliance or ↑ requirements due to ↑ stress (eg, infection) → lipolysis and oxidation of free fatty acids → ↑ ketone bodies (β -hydroxybutyrate > acetoacetate). Insulin deficient, ketones present.	Profound hyperglycemia → excessive osmotic diuresis → dehydration and ↑ serum osmolality → HHS. Classically seen in older patients with type 2 DM and limited ability to drink. Insulin present, ketones deficient.
SIGNS/SYMPTOMS	DKA is D eadly: D elirium/psychosis, K ussmaul respirations (rapid, deep breathing), A bdominal pain/nausea/vomiting, D ehydration. Fruity breath odor due to exhaled acetone.	Thirst, polyuria, lethargy, focal neurologic deficits, seizures.
LABS	Hyperglycemia, ↑ H^+ , ↓ HCO_3^- (↑ anion gap metabolic acidosis), ↑ urine and blood ketone levels, leukocytosis. Normal/↑ serum K^+ , but depleted intracellular K^+ due to transcellular shift from ↓ insulin and acidosis. Osmotic diuresis → ↑ K^+ loss in urine → total body K^+ depletion.	Hyperglycemia (often > 600 mg/dL), ↑ serum osmolality (> 320 mOsm/kg), normal pH (no acidosis), no ketones. Normal/↑ serum K^+ , ↓ intracellular K^+ .
COMPLICATIONS	Life-threatening mucormycosis, cerebral edema, cardiac arrhythmias.	Can progress to coma and death if untreated.
TREATMENT	IV fluids, IV insulin, and K^+ (to replete intracellular stores). Glucose may be required to prevent hypoglycemia from insulin therapy.	

Hypoglycemia in diabetes mellitus

Usually occurs in patients treated with insulin or insulin secretagogues (eg, sulfonylureas, meglitinides) in the setting of high-dose treatment, inadequate food intake, and/or exercise.

- Neurogenic (autonomic) symptoms: diaphoresis, tachycardia, tremor, anxiety, hunger. Allow perception of ↓ glucose (hypoglycemia awareness).
- Neuroglycopenic symptoms: altered mental status, seizures, death due to insufficient glucose in CNS. May occur in the absence of preceding neurogenic symptoms in patients with attenuated autonomic response, eg, β -blocker use (ie, hypoglycemia unawareness).

Treatment: simple carbohydrates (eg, glucose tablets, fruit juice), IM glucagon, IV dextrose.

Cushing syndrome

ETIOLOGY

↑ cortisol due to a variety of causes:

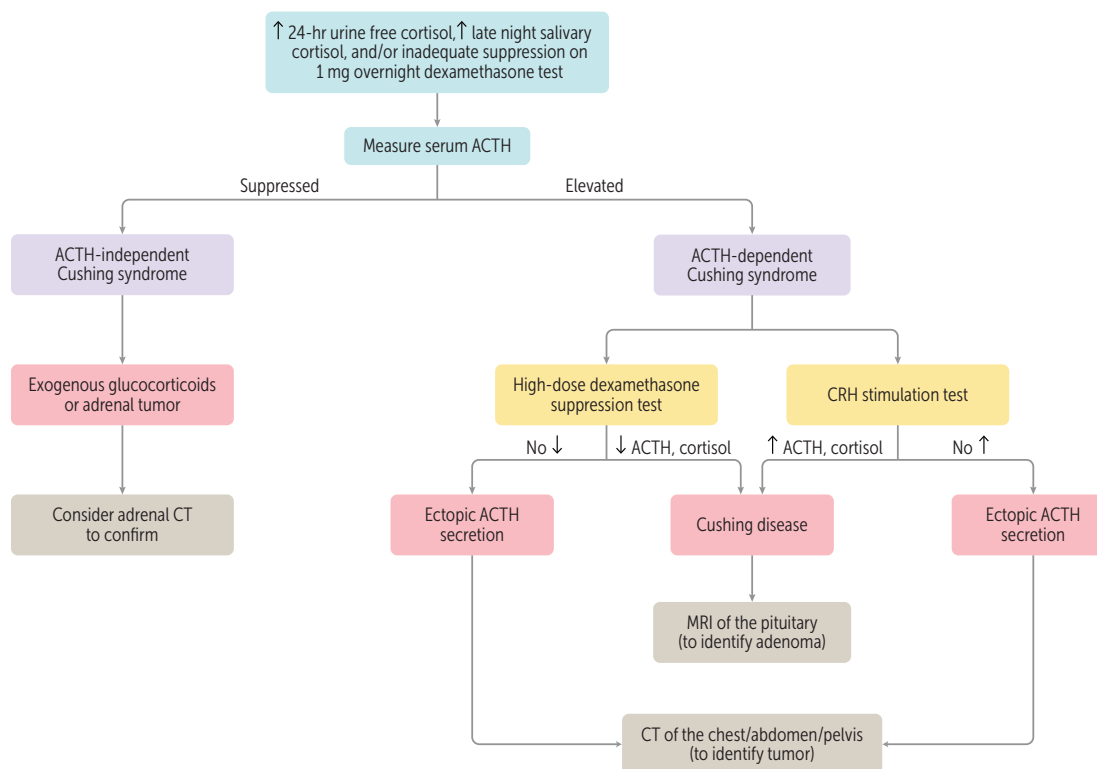
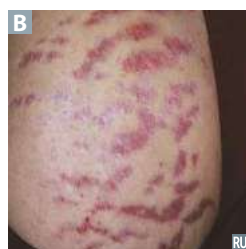
- Exogenous glucocorticoids → ↓ ACTH → bilateral adrenal atrophy. Most common cause.
- Primary adrenal adenoma, hyperplasia, or carcinoma → ↓ ACTH → atrophy of uninvolved adrenal gland.
- ACTH-secreting pituitary adenoma (Cushing disease); paraneoplastic ACTH secretion (eg, small cell lung cancer, bronchial carcinoids) → bilateral adrenal hyperplasia. Cushing disease is responsible for the majority of endogenous cases of Cushing syndrome.

FINDINGS

MOON FACIES: Metabolic syndrome (hypertension, hyperglycemia, hyperlipidemia), **O**besity (truncal weight gain with wasting of extremities, round “moon” facies **A**, dorsocervical fat pad “buffalo hump”), **O**steoporosis, **N**europsychiatric (depression, anxiety, irritability), **F**acial plethora, **A**ndrogen excess (acne, hirsutism), **C**ataract, **I**mmunosuppression, **E**cchymoses (easy bruising), **S**kin changes (thinning, striae **B**, hyperpigmentation).

DIAGNOSIS

Screening tests include: ↑ free cortisol on 24-hr urinalysis, ↑ late night salivary cortisol, and no suppression with overnight low-dose dexamethasone test.



Adrenal insufficiency

Inability of adrenal glands to generate enough glucocorticoids +/- mineralocorticoids for the body's needs. Can be acute or chronic. Symptoms include weakness, fatigue, orthostatic hypotension, muscle aches, weight loss, GI disturbances, sugar and/or salt cravings.

Treatment: glucocorticoid +/- mineralocorticoid replacement.

Primary adrenal insufficiency

↓ gland function → ↓ cortisol, ↓ aldosterone → hypotension (hyponatremic volume contraction), hyperkalemia, metabolic acidosis, skin/mucosal (pressure areas and buccal mucosa) hyperpigmentation **A**. ↓ cortisol → ↓ negative feedback effect on the pituitary → ↑ production of POMC derivatives (β-endorphin, ACTH, MSH) → ↑ melanin synthesis due to ↑ MSH. **P**rimarily pigments the skin/mucosa.

Addison disease—chronic 1° adrenal insufficiency; caused by adrenal atrophy or destruction. Most commonly due to autoimmune adrenalitis (high-income countries) or TB (low-income countries).

Secondary and tertiary adrenal insufficiency

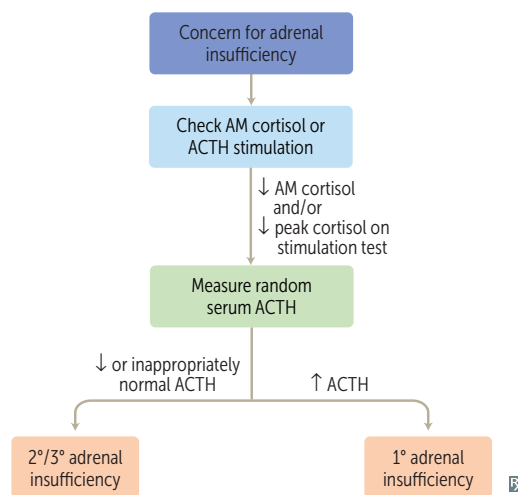
↓ pituitary ACTH secretion (secondary) or ↓ hypothalamic CRH secretion (tertiary). No hyperkalemia (aldosterone synthesis preserved due to functioning adrenal gland, intact RAAS), no hyperpigmentation.

2° adrenal insufficiency is due to pituitary pathologies, 3° adrenal insufficiency is most commonly due to abrupt cessation of chronic glucocorticoid therapy (HPA suppression). **T**ertiary from treatment.

Acute adrenal insufficiency

Also called adrenal (addisonian) crisis; often precipitated by acute stressors that ↑ glucocorticoid requirements (eg, infection) in patients with pre-existing adrenal insufficiency or on glucocorticoid therapy. May present with acute abdominal pain, nausea, vomiting, altered mental status, shock.

Waterhouse-Friderichsen syndrome—bilateral adrenal hemorrhage in the setting of sepsis (eg, meningococemia) → acute 1° adrenal insufficiency.



Hyperaldosteronism

Increased secretion of aldosterone from adrenal gland. Clinical features include hypertension, ↓ or normal K^+ , metabolic alkalosis. 1° hyperaldosteronism does not directly cause edema due to aldosterone escape mechanism. However, certain 2° causes of hyperaldosteronism (eg, heart failure) impair the aldosterone escape mechanism, leading to worsening of edema.

Primary hyperaldosteronism

Seen in patients with bilateral adrenal hyperplasia or adrenal adenoma (Conn syndrome). ↑ aldosterone, ↓ renin. Leads to treatment-resistant hypertension.

Secondary hyperaldosteronism

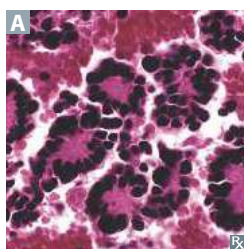
Seen in patients with renovascular hypertension, juxtaglomerular cell tumors (renin-producing), and edema (eg, cirrhosis, heart failure, nephrotic syndrome). ↑ aldosterone, ↑ renin.

Neuroendocrine tumors

Heterogeneous group of neoplasms originating from neuroendocrine cells (which have traits similar to nerve cells and hormone-producing cells).

Most neoplasms occur in the GI system (eg, carcinoid, gastrinoma), pancreas (eg, insulinoma, glucagonoma), and lungs (eg, small cell carcinoma). Also in thyroid (eg, medullary carcinoma) and adrenals (eg, pheochromocytoma).

Neuroendocrine cells (eg, pancreatic β cells, enterochromaffin cells) share a common biologic function through amine precursor uptake decarboxylase (APUD) despite differences in embryologic origin, anatomic site, and secretory products (eg, chromogranin A, neuron-specific enolase [NSE], synaptophysin, serotonin, histamine, calcitonin). Treatment: surgical resection, somatostatin analogs.

Neuroblastoma

Most common solid extracranial tumor in children (typically < 4 years old). Usually arises in adrenal medulla, but may occur anywhere along the sympathetic chain. Originates from neural crest cells.

Most common presentation is abdominal distension and a firm, irregular mass that can cross the midline (vs Wilms tumor, which is smooth and unilateral). Less likely to develop hypertension than with pheochromocytoma (neuroblastoma is normotensive). Can also present with opsoclonus-myoclonus syndrome (“dancing eyes-dancing feet”).

↑ HVA and VMA (catecholamine metabolites) in urine. Homer-Wright rosettes (neuroblasts surrounding a central area of neuropil **A**) characteristic of neuroblastoma and medulloblastoma. Bombesin and NSE ⊕. Associated with amplification of *N-myc* oncogene.

Pheochromocytoma**ETIOLOGY**

Most common tumor of the adrenal medulla in adults (black arrow in **A**; red arrow points to bone metastases). Derived from chromaffin cells (arise from neural crest). Rare.

May be associated with germline mutations (eg, *NF-1*, *VHL*, *RET* [MEN 2A, 2B]).

Rule of 10's:

10% malignant

10% bilateral

10% extra-adrenal (paraganglioma; eg, bladder wall, organ of Zuckerkandl)

10% calcify

10% kids

SYMPTOMS

Most tumors secrete epinephrine, norepinephrine, and dopamine, which can cause episodic hypertension. May also secrete EPO → polycythemia.

Symptoms occur in “spells”—relapse and remit.

Episodic hyperadrenergic symptoms (**5 P's**):

Pressure (↑ BP)

Pain (headache)

Perspiration

Palpitations (tachycardia)

Pallor

FINDINGS

↑ catecholamines and metanephrines (eg, homovanillic acid, vanillylmandelic acid) in urine and plasma; episodic hyperglycemia may be an uncommon finding.

Chromogranin, synaptophysin and NSE ⊕.

TREATMENT

Irreversible α -antagonists (eg, phenoxybenzamine) followed by β -blockers prior to tumor resection. α -blockade must be achieved before giving β -blockers to avoid a hypertensive crisis. **A** before **B**.

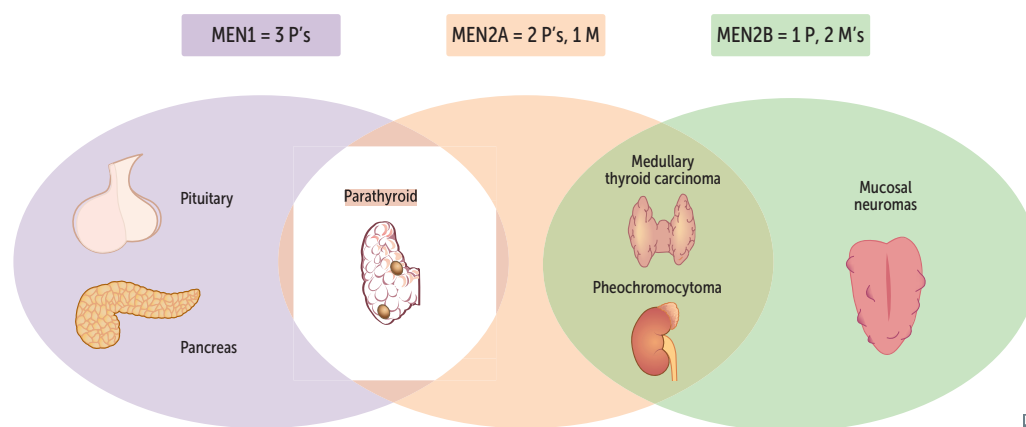
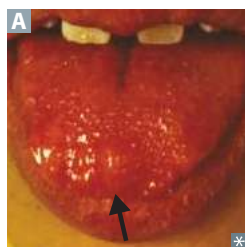
Phenoxybenzamine for **p**heochromocytoma.

Multiple endocrine neoplasias

All **MEN** syndromes have autosomal **dominant** inheritance.

The X-**MEN** are **dominant** over villains.

SUBTYPE	CHARACTERISTICS
MEN1	Pituitary tumors (prolactin or GH) Pancreatic endocrine tumors—Zollinger-Ellison syndrome, insulinomas, VIPomas, glucagonomas (rare) Parathyroid adenomas Associated with mutation of <i>MEN1</i> (tumor suppressor, codes for menin, chromosome 11), angiofibromas, collagenomas, meningiomas
MEN2A	Parathyroid hyperplasia Medullary thyroid carcinoma—neoplasm of parafollicular C cells; secretes calcitonin; prophylactic thyroidectomy required Pheochromocytoma (secretes catecholamines) Associated with mutation in <i>RET</i> (protooncogene, codes for receptor tyrosine kinase, chromosome 10)
MEN2B	Medullary thyroid carcinoma Pheochromocytoma Mucosal neuromas A (oral/intestinal ganglioneuromatosis) Associated with marfanoid habitus; mutation in <i>RET</i> gene



Pancreatic islet cell tumors**Insulinoma**

Tumor of pancreatic β cells $\rightarrow$ overproduction of insulin $\rightarrow$ hypoglycemia.
 May see Whipple triad: low blood glucose, symptoms of hypoglycemia (eg, lethargy, syncope, diplopia), and resolution of symptoms after normalization of plasma glucose levels. Symptomatic patients have $\downarrow$ blood glucose and $\uparrow$ C-peptide levels (vs exogenous insulin use). $\sim 10\%$ of cases associated with MEN1 syndrome.
 Treatment: surgical resection.

Glucagonoma

Tumor of pancreatic α cells $\rightarrow$ overproduction of glucagon.
 Presents with **6 D's**: **d**ermatitis (necrolytic migratory erythema), **d**iabetes (hyperglycemia), **DVT**, **d**eclining weight, **d**epression, **d**iarrhea.
 Treatment: octreotide, surgical resection.

Somatostatinoma

Tumor of pancreatic δ cells $\rightarrow$ overproduction of somatostatin $\rightarrow$ $\downarrow$ secretion of secretin, cholecystokinin, glucagon, insulin, gastrin, gastric inhibitory peptide (GIP).
 May present with diabetes/glucose intolerance, steatorrhea, gallstones, achlorhydria.
 Treatment: surgical resection; somatostatin analogs (eg, octreotide) for symptom control.

Carcinoid tumors**A**

Carcinoid tumors arise from neuroendocrine cells, most commonly in the intestine or lung. Neuroendocrine cells secrete 5-HT, which undergoes hepatic first-pass metabolism and enzymatic breakdown by MAO in the lung. If 5-HT reaches the systemic circulation (eg, after liver metastasis), carcinoid tumor may present with **carcinoid syndrome**—episodic flushing, diarrhea, wheezing, right-sided valvular heart disease (eg, tricuspid regurgitation, pulmonic stenosis), $\uparrow$ urinary 5-HIAA. Excess 5-HT production depletes tryptophan stores, leading to niacin deficiency (pellagra).

✱

Histology: rosettes **A**, chromogranin A $\oplus$, synaptophysin $\oplus$.
 Diagnosis: $\uparrow$ 24-hour urinary excretion of 5-HIAA. Octreotide scintigraphy $\rightarrow$ tumor localization and staging.
 Treatment: surgical resection, somatostatin analog (eg, octreotide) or tryptophan hydroxylase inhibitor (eg, telotristat) for symptom control.

Rule of thirds:

- 1/3** metastasize
- 1/3** present with 2nd malignancy
- 1/3** are multiple

Zollinger-Ellison syndrome

Constellation of symptoms resulting in acid hypersecretion from gastrin-secreting tumor (gastrinomas) in duodenum (most common) or pancreas $\rightarrow$ single or multiple, recurrent ulcers in the duodenum/jejunum (often refractory to proton pump inhibitors) with resulting malabsorption (acid inactivation of pancreatic enzymes secreted into the small bowel). Presents with abdominal pain, heartburn, diarrhea/steatorrhea, weight loss. Positive secretin stimulation test ($\uparrow$ gastrin levels after secretin administration, which normally inhibits gastrin release). Chromogranin A $\oplus$. May be associated with MEN1.